

AI Practice Exams

Practice Exam 1

Suggested time: 2 hours. Total: 100 points. Closed notes; calculator allowed. Use the logistic function $\varphi(v) = \frac{1}{1 + e^{-v}}$ (slope parameter $a = 1$) unless stated. Round to four decimal places.

Part I. Identification (20 points, 1 each)

1. The approach to AI that judges intelligence by whether a system acts like a human.
2. The 1956 event widely considered the founding event of AI.
3. The decision-making strategy that aims for a satisfactory or adequate result instead of the optimal solution.
4. The first commercial expert system (McDermott, 1982).
5. The law of intelligent action illustrated by Deep Blue computing 200 million moves per second with a small knowledge base.
6. Complete the equation: $\text{Agent} = \text{_____} + \text{_____}$.
7. The four items that PEAS stands for.
8. The learning-agent component that suggests exploratory actions leading to new and informative experiences.
9. The type of environment that does not change with the passage of time while the agent's performance score does.
10. The collection of nodes waiting to be expanded.
11. The blind search strategy that calls depth-limited search with increasing limits until a goal is found.
12. The relation between h_2 and h_1 when $h_2(n) \geq h_1(n)$ for every node n .
13. The name of the equation $f(n') = \max(f(n), g(n') + h(n'))$.
14. The problem in game search where a useless move pushes an unavoidable threat beyond the search depth.
15. The four steps of each Monte Carlo Tree Search iteration, in order.
16. In a CSP, the name for an assignment that is both consistent and complete.
17. The variant of the perceptron algorithm that converges to an optimal solution even when the classes are not linearly separable.
18. The metric defined as the harmonic mean of precision and recall.
19. The junction point of two neurons across which a nerve impulse passes.
20. The name of the quantity $\delta_j(n)$ in the backpropagation algorithm.

Part II. True or False (20 points, 1 each)

Write TRUE or FALSE. If false, state the correction in one line.

1. The Turing Test approach places its emphasis on thinking instead of behavior.
2. A rational agent maximizes actual performance.
3. Backgammon is a fully observable, stochastic environment.
4. Breadth-first search is optimal for any assignment of step costs.
5. Depth-first search has space complexity $O(bm)$.
6. Iterative deepening search has space complexity $O(b^d)$.
7. Greedy search is complete but not optimal.
8. Every admissible heuristic is also consistent (monotonic).

9. A* with $h(n) = 0$ for all n behaves like uniform-cost search.
10. Alpha-beta pruning can change the minimax value computed at the root.
11. With perfect move ordering, alpha-beta has time complexity $O(b^{m/2})$.
12. Forward checking detects every failure that arc consistency detects.
13. A CSP whose constraint graph has no loops can be solved in $O(nd^2)$ time.
14. The perceptron algorithm converges for any training set.
15. Sensitivity and recall are the same quantity.
16. For highly unbalanced data, accuracy is a reliable measure of classifier quality.
17. Hebb proposed the first learning rule in 1949.
18. A two-layer perceptron can separate classes formed by any arbitrary union of polyhedral regions.
19. In batch mode, weights are updated after the presentation of each training example.
20. Backpropagation is guaranteed to converge to the global minimum of the error surface.

Part III. Proving and Derivation (20 points)

1. (5 points) Prove that A* search using an admissible heuristic never returns a suboptimal goal.
2. (5 points) For the 8-puzzle, show that the Manhattan distance heuristic h_2 is (a) admissible and (b) consistent.
3. (10 points) **Full derivation of the backpropagation algorithm.** Consider a multilayer perceptron in which neuron i feeds neuron j , and neuron j feeds neuron k . Let

$$v_j(n) = \sum_{i=0}^p w_{ji}(n) y_i(n), \quad y_j(n) = \varphi(v_j(n)), \quad e_j(n) = d_j(n) - y_j(n), \quad \mathcal{E}(n) = \frac{1}{2} \sum_{j \in C} e_j^2(n).$$

Starting from $\Delta w_{ji}(n) = -\eta \frac{\partial \mathcal{E}(n)}{\partial w_{ji}(n)}$:

- (a) Derive $\Delta w_{ji}(n) = \eta \delta_j(n) y_i(n)$ for an *output* neuron j and give $\delta_j(n)$. Show every factor of the chain rule.
- (b) Derive $\delta_j(n)$ for a *hidden* neuron j . State why the result of (a) cannot be used directly.
- (c) For the logistic function $\varphi(v) = \frac{1}{1 + e^{-av}}$, derive $\varphi'(v)$ in terms of $\varphi(v)$, and write the final expressions of $\delta_j(n)$ for an output neuron and for a hidden neuron.

Part IV. Calculation (30 points)

Problem 1. A* search (7 points). Undirected graph with step costs:

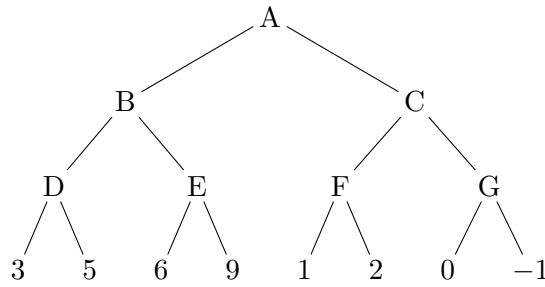
Edge	S-A	S-B	A-B	A-C	B-C	B-D	C-D	C-G	D-G
Cost	2	4	1	5	3	6	1	4	2

Node n	S	A	B	C	D	G
$h(n)$	8	7	6	2	2	0

Start S, goal G. Break ties alphabetically. Do not revisit a node already on the current path.

- (a) Compute $h^*(n)$ for every node. Is h admissible?
- (b) Is h consistent? If not, name an edge that violates the condition.
- (c) Trace A*: list each node expanded with its g and f , and give the solution path and cost.
- (d) Give the path and cost returned by greedy search.

Problem 2. Alpha-beta pruning (5 points). The root A is a MAX node. Levels alternate MAX, MIN, MAX; the leaves are terminal values.



- Compute the minimax value of every internal node.
- Apply alpha-beta pruning left to right. List the leaves that are evaluated and the leaves that are pruned, with the condition that triggers each prune.

Problem 3. Perceptron (5 points). Use the one-sample-at-a-time perceptron algorithm with $\rho = 0.5$ on extended vectors $\mathbf{x} = [x_1, x_2, 1]^T$. Initial weight vector $\mathbf{w}(0) = [1, -1, 0]^T$. Present the samples in this order, repeating until an epoch produces no update:

Sample	(x_1, x_2)	Class
1	(2, 2)	ω_1
2	(1, -1)	ω_2
3	(-1, 1)	ω_2
4	(3, 1)	ω_1

Update rule:

$$\mathbf{w}(t+1) = \begin{cases} \mathbf{w}(t) + \rho \mathbf{x}_{(t)} & \text{if } \mathbf{x}_{(t)} \in \omega_1 \text{ and } \mathbf{w}^T(t) \mathbf{x}_{(t)} \leq 0 \\ \mathbf{w}(t) - \rho \mathbf{x}_{(t)} & \text{if } \mathbf{x}_{(t)} \in \omega_2 \text{ and } \mathbf{w}^T(t) \mathbf{x}_{(t)} \geq 0 \\ \mathbf{w}(t) & \text{otherwise} \end{cases}$$

Give the weight vector after every update and the final decision line.

Problem 4. Classifier evaluation (5 points). A disease classifier gives $TP = 40$, $FN = 10$, $FP = 20$, $TN = 130$. Compute accuracy, precision, recall, specificity, F_1 , balanced accuracy, and MCC.

Problem 5. Backpropagation (6 points). Network 2–2–1, logistic activations, bias input +1, $\eta = 1$. Input $\mathbf{x} = (1, 1)$, target $d = 0$.

Neuron	Bias	First weight	Second weight
Hidden h_1	0.0	0.5 (from x_1)	-0.5 (from x_2)
Hidden h_2	0.2	0.3 (from x_1)	0.1 (from x_2)
Output o	-0.1	0.8 (from h_1)	0.4 (from h_2)

- Forward pass: y_{h_1} , y_{h_2} , o , and $\mathcal{E}$.
- Local gradients δ_o , δ_{h_1} , δ_{h_2} .
- All nine updated weights after one iteration.

Problem 6. Linear classifier geometry and gradient descent (2 points).

- (a) For $g(\mathbf{x}) = 3x_1 + 4x_2 - 12$, find the distance of the decision line from the origin and the distance of the point $(2, 1)$ from the line. Which side of the line is $(2, 1)$ on?
- (b) For $J(\theta) = (\theta - 3)^2$ with $\mu = 0.25$ and $\theta(0) = 0$, compute $\theta(1)$, $\theta(2)$, $\theta(3)$. For what range of μ does the iteration converge?

Part V. Essay (10 points; answer any two, 5 each)

1. Compare the simple reflex, model-based reflex, goal-based, and utility-based agents. Which is the most appropriate design for an automated taxi, and why? Refer to the properties of the taxi's task environment.
2. Explain the laws "Search compensates for lack of knowledge" and "Knowledge compensates for lack of search". Give one example of each and relate both laws to the role of the heuristic function in A^* .
3. Explain why the XOR problem was a turning point for neural networks, how the multilayer perceptron with backpropagation resolves it, and two limitations that remain.

Practice Exam 2

Suggested time: 2 hours. Total: 100 points. Same conventions as Exam 1. This exam leans more on blind search, stochastic games, CSPs, LMS, and multi-output backpropagation.

Part I. Identification (20 points, 1 each)

1. The approach to AI based on Aristotelian logic and syllogisms.
2. The person who proposed bounded (limited) rationality, and the year.
3. The “magic number” of chunks of information an expert knows.
4. An algorithm that can be interrupted at any time and returns results whose value monotonically increases with time.
5. A function that maps a state onto a real number and provides the trade-off between conflicting goals.
6. The property of a system whose behavior is determined by its own experience.
7. The problem type in which the agent is ignorant of the effects of its actions and must experiment.
8. The blind search strategy, attributed to Dijkstra (1959), that always expands the lowest path-cost node on the fringe.
9. A heuristic that never overestimates the cost to reach the goal.
10. Korf’s modification of A* that reduces space complexity from exponential to linear.
11. The quantity b^* defined by $N = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$.
12. In alpha-beta search, the value of the best (highest) choice found so far along the path for MAX.
13. The algorithm that extends minimax to games with chance nodes.
14. Positions in which the evaluation function value is changing slowly from ply to ply.
15. The standard algorithm that enforces arc consistency using a queue of arcs.
16. The technique of instantiating a set of variables, in all ways, so that the remaining constraint graph is a tree.
17. The linear neuron trained with the LMS algorithm, introduced by Widrow and Hoff.
18. The matrix $R_x = E[\mathbf{x}\mathbf{x}^T]$.
19. The metric $\frac{TN}{TN + FP}$.
20. A neural network that has at least one feedback loop.

Part II. True or False (20 points, 1 each)

Write TRUE or FALSE. If false, state the correction in one line.

1. AIMA defines AI as the study and construction of rational agents.
2. The Lighthill Report (1973) contributed to the AI Winter.
3. A simple reflex agent selects actions on the basis of the entire percept history.
4. Chess played with a clock is a static environment.
5. A crossword puzzle is fully observable, single-agent, deterministic, sequential, static, and discrete.
6. Uniform-cost search finds the cheapest solution provided the cost of a path never decreases as we go along the path.
7. Depth-limited search is complete if the depth limit is at least the depth of the solution.
8. Bidirectional search has time complexity $O(b^{d/2})$.
9. Computation time, more than memory, is the main drawback of A*.
10. If h_1 and h_2 are admissible, then $\max(h_1, h_2)$ is admissible.
11. The minimax decision assumes the opponent plays perfectly.
12. MCTS returns the move whose node has the highest win percentage.
13. In expectiminimax, any order-preserving transformation of the leaf values leaves the chosen move

unchanged.

14. To make the arc $X \rightarrow Y$ consistent, values are deleted from the domain of Y .
15. Backtracking search for CSPs is breadth-first search with single-variable assignments.
16. The LMS algorithm converges asymptotically to the MSE solution.
17. A model with many parameters relative to the training set size tends to have high bias and low variance.
18. PPV and NPV depend on the proportion of classes in the test set.
19. The slope of the logistic function $\varphi(v) = \frac{1}{1 + e^{-av}}$ at the origin is $a/4$.
20. With momentum, when the gradient keeps the same sign on consecutive iterations, the weight change decreases in magnitude.

Part III. Proving and Derivation (20 points)

1. (5 points) A heuristic is consistent if $h(n) \leq c(n, n') + h(n')$ for every successor n' of n , with $h(\text{goal}) = 0$. Prove that (a) every consistent heuristic is admissible and (b) $f(n) = g(n) + h(n)$ is non-decreasing along any path.
2. (5 points) (a) Prove that no single perceptron with a step activation can compute XOR. (b) Give weights for a two-layer perceptron that computes XOR and verify all four inputs.
3. (10 points) **Full derivation of the backpropagation algorithm with the hyperbolic tangent activation.** Use the notation

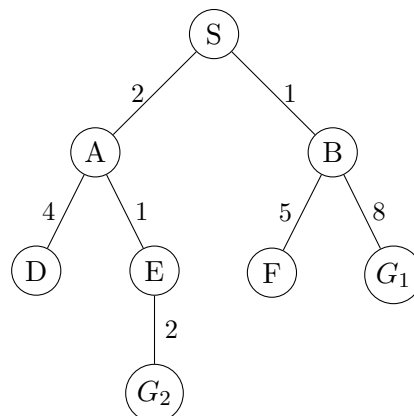
$$v_j(n) = \sum_{i=0}^p w_{ji}(n) y_i(n), \quad y_j(n) = \varphi(v_j(n)), \quad e_j(n) = d_j(n) - y_j(n), \quad \mathcal{E}(n) = \frac{1}{2} \sum_{j \in C} e_j^2(n),$$

and define the local gradient as $\delta_j(n) = -\frac{\partial \mathcal{E}(n)}{\partial v_j(n)}$.

- (a) Show that $\Delta w_{ji}(n) = \eta \delta_j(n) y_i(n)$ holds for every neuron, output or hidden.
- (b) Derive $\delta_j(n)$ for an output neuron and for a hidden neuron.
- (c) For $\varphi(v) = a \tanh(bv)$, derive $\varphi'(v)$ in terms of $\varphi(v)$ and give the final expressions of $\delta_j(n)$ for both cases.
- (d) Write the generalized delta rule with momentum constant α , and explain the effect of the momentum term when $\frac{\partial \mathcal{E}(n)}{\partial w_{ji}(n)}$ has (i) the same sign and (ii) opposite signs on consecutive iterations.

Part IV. Calculation (30 points)

Problem 1. Blind search (5 points). Search tree with edge costs on the edges. G_1 and G_2 are goals. Children are generated left to right.



For each of BFS, DFS, UCS, and IDS: give the order in which nodes are expanded (goal test when a

node is selected), the goal found, and the path cost. Which strategies returned the optimal solution, and was that guaranteed?

Problem 2. Game trees (5 points).

- (a) MAX root with three MIN children; leaves left to right are (5, 7, 3), (4, 1, 8), (6, 2, 9). Give the minimax value and the leaves pruned by alpha-beta.
- (b) MAX chooses between moves a_1 and a_2 , each leading to a chance node whose children are MIN nodes:

Move	Outcome 1	Outcome 2
a_1	probability 0.5, MIN over leaves (6, 10)	probability 0.5, MIN over leaves (2, 12)
a_2	probability 0.25, MIN over leaves (20, 8)	probability 0.75, MIN over leaves (4, 5)

Compute the expectiminimax value of each move and state MAX's choice.

Problem 3. Constraint satisfaction (5 points). Variables A, B, C, D, E , each with domain $\{R, G, B\}$. Constraints: $A \neq B, A \neq C, B \neq C, B \neq D, C \neq D, D \neq E$.

- (a) Assign $A = R$. Show all domains after forward checking.
- (b) Then assign $B = G$. Show all domains after forward checking.
- (c) Starting from (b), enforce arc consistency. State each deletion and the arc that caused it.
- (d) Is the constraint graph a tree? Give a smallest cycle cutset and the resulting worst-case runtime bound.

Problem 4. Multi-class evaluation (4 points). Rows are the true class, columns the predicted class.

True \ Predicted	X	Y	Z
X	30	5	5
Y	10	40	10
Z	0	10	90

Compute the accuracy, the recall and precision of each class, the TP , FP , FN , TN counts for class Y, and the balanced accuracy.

Problem 5. LMS (4 points). Start with $\mathbf{w}(0) = [0, 0, 0]^T$ on extended vectors $[x_1, x_2, 1]^T$ and $\rho = 0.1$. Apply

$$\mathbf{w}(k) = \mathbf{w}(k-1) + \rho \mathbf{x}_k (y_k - \mathbf{x}_k^T \mathbf{w}(k-1))$$

to the samples in order:

k	(x_1, x_2)	y
1	(1, 2)	+1
2	(2, -1)	-1
3	(0, 1)	+1

Give $\mathbf{w}$ after each step. State one difference between the LMS update and the perceptron update.

Problem 6. Backpropagation with two outputs (7 points). Network 2–2–2, logistic activations, bias input +1, $\eta = 0.5$. Input $\mathbf{x} = (0.5, 1.0)$, target $\mathbf{d} = (1, 0)$.

Neuron	Bias	First weight	Second weight
Hidden h_1	0.1	0.2 (from x_1)	0.4 (from x_2)
Hidden h_2	−0.2	0.6 (from x_1)	−0.1 (from x_2)
Output o_1	0.3	0.5 (from h_1)	−0.6 (from h_2)
Output o_2	−0.1	0.7 (from h_1)	0.2 (from h_2)

- Forward pass: y_{h_1} , y_{h_2} , o_1 , o_2 , and $\mathcal{E}$.
- δ for both output neurons and both hidden neurons.
- Updated weights of o_1 (3 values) and of h_1 (3 values).
- Suppose the previous change of the weight from h_1 to o_1 was 0.02 and the momentum constant is $\alpha = 0.9$. Compute the new Δw for that weight.

Part V. Essay (10 points; answer any two, 5 each)

- Compare breadth-first search, depth-first search, iterative deepening search, and A* on completeness, optimality, time, and space. Which would you use for a large search space with unknown solution depth and no heuristic, and why?
- A hospital reports that its disease classifier is 97% accurate. Explain why this number alone is not enough, which metrics you would ask for, and how the decision threshold affects them.
- Explain the bias-variance dilemma and relate it to generalization and overfitting in a multilayer perceptron. Name three practices from the lecture that improve training or generalization.